

Generation Interconnection System Impact Study Report

For Queue Project Q18-0988

Garnet Mesa 138 kV

March 01, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0988. The Interconnection Customer proposes a 156.7 MW solar photovoltaic facility interconnecting at the Garnet Mesa 138 kV substation in the Pennsylvania area. The request was received on May 07, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 299 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,758,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Moweaqua / Garnet Mesa 138 kV	M. Vernon / Garnet Mesa 138 kV	184.9	156	100.0
Columbus-Rapelje Auto / Sugar Hill 138 kV	Ritter Park / Sugar Hill 345 kV	132.9	231	44.0
Columbus-Rapelje Auto / Chadwicks 138 kV line	Sugar Hill / Ritter Park 345 kV Line	117.4	271	44.0
Chadwicks / Buffalo Rd 138 kV Line	Ritter Park / Sugar Hill 345 kV	116.0	284	44.0
Buffalo Rd / Mt. Vernon 138 kV	Ritter Park / Sugar Hill 345 kV	115.5	290	44.0
Mt. Vernon / Garnet Mesa 138 kV	Ritter Park / Sugar Hill 345 kV	114.5	300	44.0
Marys Lake Powerplant / United Cogen Inc 138 kV	Sugar Hill / Allegheny Energy Units 3 4 & 5 345 kV L	110.4	203	8.8
Sugar Hill Substation / 28Th St 138 kV	Sugar Hill / Allegheny Energy Units 3 4 & 5 345 kV	110.3	255	8.8
Pine Crk / 28Th St 138 kV	Allegheny Energy Units 3 4 & 5 / Sugar Hill 345 kV	109.1	251	8.8
Swif 1 / Marys Lake Powerplant 138 kV L	Allegheny Energy Units 3 4 & 5 / Sugar Hill 345 kV	108.7	216	8.8
Swift 1 / Pine Crk 138 kV Line	Sugar Hill / Allegheny Energy Units 3 4 & 5 345 kV	108.2	243	8.8
Allegheny No. 5 Hydro Sta / Stinson 138 kV	Regal / Yucca 138 kV	102.1	222	5.9
Siletz / Grand Haven Diesel Plant 138 kV	Sugar Hill / Allegheny Energy Units 3 4 & 5 345 kV	101.9	195	8.8
United Cogen Inc / Siletz 138 kV	Sugar Hill / Allegheny Energy Units 3 4 & 5 345 kV	101.2	204	8.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lions Rd 345-138 kV Autotransformer	Garnet Mesa / Mt. Vernon 138 kV	100.3	381	7.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Moweaqua / Garnet Mesa 138 kV	Rebuild line with higher-capacity conductor	\$10,657,000	56% (\$5,956,000)
Columbus-Rapelje Auto / Sugar Hill 138 kV	Rebuild line with higher-capacity conductor	\$2,118,000	44% (\$927,000)
Columbus-Rapelje Auto / Chadwicks 138 kV line	Rebuild line with higher-capacity conductor	\$14,638,000	44% (\$6,406,000)
Buffalo Rd / Chadwicks 138 kV line	Rebuild line with higher-capacity conductor	\$5,125,000	44% (\$2,243,000)
Buffalo Rd / Mt. Vernon 138 kV	Rebuild line with higher-capacity conductor	\$4,673,000	44% (\$2,045,000)
Mt. Vernon / Garnet Mesa 138 kV	Rebuild line with higher-capacity conductor	\$3,595,000	44% (\$1,574,000)
Marys Lake Powerplant / United Cogen Inc 138 kV	Rebuild line with higher-capacity conductor	\$5,186,000	15% (\$800,000)
28Th St / Sugar Hill 138 kV	Rebuild line with higher-capacity conductor	\$1,050,000	15% (\$162,000)
28Th St / Pine Creek 138 kV	Reconductor line	\$1,082,000	15% (\$167,000)
Swift 1 Substation / Marys Lake Powerplant 138 kV	Reconductor line	\$360,000	15% (\$56,000)
Swift 1 / Pine Crk 138 kV Line	Reconductor line	\$312,000	15% (\$48,000)
Allegheny No. 5 Hydro Sta / Stinson 138 kV Line	Reconductor line	\$1,198,000	17% (\$202,000)
Grand Haven Diesel Plant Substation / Siletz 138 kV	Reconductor line	\$1,070,000	15% (\$165,000)
United Cogen Inc / Siletz 138 kV	Reconductor line	\$358,000	15% (\$55,000)
Lions Rd 345/138kV Transformer	Replace transformer with higher-rated unit	\$17,868,000	56% (\$9,952,000)
	Total allocated to project		\$30,758,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.